Saurav Dutta

Contact Information

Laboratoire Navier
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Educational Qualifications

Institut Polytechnique de Paris (IPP), France *(Sept '26 - Present)*

- Doctor of Philosophy (PhD) in Mechanical Sciences & Engineering

National Institute of Technology, Silchar, India *(June '19 - June'23)*

- Bachelor of Technology (B.Tech.) Civil Engineering

Publications/ Book Chapters/ Conferences

1. **S. Dutta**, V. Anand (2024). Framework for Ground Motion Characterization. *Seismic Hazard Analyses, Wave Propagation & Site Characterization, Springer Nature*
2. A. Joshi, **S. Dutta**, S. Kumar (2025, July). Hetero–EUCLID : Simultaneously segmenting and discovering hyperelastic constitutive models of all components of a heterogeneous hyperelastic material using EUCLID. *European Solid Mechanics Conference, Lyon, France.*
3. K. L. Chaurasiya, **S. Dutta**, S. Kumar, A. Joshi (2026). Hetero-EUCLID : Interpretable model discovery for heterogeneous hyperelastic materials using stress-unsupervised learning. *Computer Methods in Applied Mechanics & Engineering.*
4. H. K. Sandhu, **S. Dutta**, R. Chaunsali (2026). Wave propagation in an elastic lattice with non-reciprocal stiffness and engineered damping. *Journal of the Acoustical Society of America.*
5. S. Singh, M. Kumar, **S. Dutta**, V. Anand (2026). Identification of critical ground motion features for seismic fragility studies considering soil-structure interaction. *Journal of Vibration Engineering & Technologies.*

Professional Experience

Ecole des Ponts ParisTech (ENPC), IP Paris, France *(Sept '26 - Present)*
Doctoral Researcher, Laboratoire Navier, Advisor : Prof. Jeremy Bleyer

- Developing hybrid architectures that couple classical phenomenological models with neural components to learn **dissipative, path-dependent** behaviours with hidden internal state variables.
- Generating synthetic training data from high-fidelity RVE-scale microstructural simulations.

Vizuara AI Labs, Pune, India *(Sept '25 – March '26)*
Graduate Mentor and Scientist

- Worked on the technical modules for Artificial Intelligence for Materials, delivering lectures with Python and Julia codes.
- Mentored Candidates in the SciML and GenAI bootcamps.

Indian Institute of Science, Bengaluru *(June '23 - July '25)*
Research Associate, Dept. of Mechanical Engineering, Advisor : Prof. Akshay Joshi

- Investigated automated model discovery for heterogeneous hyperelastic materials by extending the EUCLID framework, originally developed for homogeneous systems.
- Developed a Python-based probabilistic growth algorithm with interpretable priors to detect and track heterogeneous material interfaces, achieving **98–99%** segmentation accuracy.
- Implemented Bayesian-EUCLID with **2–10%** MCMC sub-sampling over 10,000–21,000 C3D6 nodes, predicting local material parameters with **1–3%** mean error.
- Validated the resulting Hetero-EUCLID method on 3D non-equi-biaxial tension tests ($\lambda_x = 1.6, \lambda_y = 2.2$), demonstrating noise robustness and interpretable constitutive model recovery.

Indian Institute of Technology BHU, Varanasi *(May '22 - July '22)*
Summer Research Intern, Dept. of Civil Engineering, Advisor : Prof. Vishwajit Anand

- Developed a MATLAB-based framework to characterize seismic records and identify seismic parameters most sensitive to structural response under earthquake loading.

- Processed 195 seismic records from the PEER Database, extracting and computing over **30+** intensity measures, including PGA, PGV, PGD, Arias intensity.
- Utilised the OpenSeismoMatlab framework to automate Fourier and elastic response spectral analyses, determining predominant periods ($T_p = 0.1 - 2.4$ s) and pseudo-accelerations ($2.9 - 13.3$ m/s^2), providing a basis for fragility assessment and parameter sensitivity evaluation.

Teaching Experience & Talks

Course Instructor : *Artificial Intelligence for Materials*, Vizuara AI Labs *(Jan '26 – Mar '26)*

Guest Lecture : *Wave Propagation AE 351A*, Indian Institute of Science, Bengaluru *(Nov '23)*

Extracurricular Key Projects

Experimental Realisation of Non-linear Pendulums in 1-DOF *(June '23 - July '24)*
Advisor : Prof. Rajesh Chaunsali, **Indian Institute of Science, Bengaluru**

- Developed an experimental framework to realize non-linear pendulums in a 1-DOF mechanical oscillator for studying non-stationary dynamic behavior.
- Designed and fabricated custom motor hubs and magnet-embedded discs, enabling tunable magnetic interactions between the 3 pendulums.
- Implemented a custom Python control framework to synchronize 3 Dynamixel motors with **<1% error** in speed and position and generate controlled stiffness modulation.
- Observed tunable stiffness modulation from the data captured using high resolution DIC (240 fps) and LDV, developing a foundation for non-linear dynamical system.

Stress Analysis in a Truss System using Finite Element Method *(Jan '23 – Apr '23)*
Course : Finite Element Method, **National Institute of Technology, Silchar**

- Performed stress analysis of a 2D plane truss using the FEM, with a 25-element, 16-node model developed in MATLAB to evaluate structural response under a 100 kN load.
- Implemented element-level stiffness formulation to compute nodal displacements and axial stresses across all members, validating the stress distribution.
- Conducted mesh-convergence analysis for an aluminum truss ($E = 70$ GPa, $A = 900$ mm^2), determining a maximum axial stress of **115 MPa** and tip deflection of **2.4 mm**.

Technical Skills

Programming Languages	: Python, Julia, HTML, CSS, C, C++
Scientific Softwares	: MATLAB, ABAQUS, AutoCAD, COMSOL, Mathematica, CATIA
Libraries & Frameworks	: PyTorch, NumPy, SciPy, scikit-learn, Matplotlib, PyVista, OpenCV, Pandas, SymPy, Abaqus Python (scripting), Seaborn
Hardwares	: Motor Control, Arduino, Sensors, U2D2, LDV, 3D Printing

Relevant Course Work

Mechanics, Structures & Design :- Engineering Mechanics • Mechanics of Materials • Structural Analysis I, II & III • Design of Concrete Structures I & II • Design of Steel Structures • Advanced Structural Analysis • Numerical Methods in Engineering • Finite Element Method • Earthquake Resistant Design of Structures • Concrete Technology

Fluids & Hydraulics :- Fluid Mechanics • Hydraulics • Surface and Ground Water Hydrology • Water Supply Engineering • Environmental Engineering

Audited Miscellaneous :- Plates & Shells[1] • Vibrations[2] • Wave Propagation in Designed Materials[3] • Quantum Mechanics-I[3] • Condensed Matter Physics-I[3]

Workshop and Labs

CISM-IISc Workshop :- Served as a volunteer, contributing to the smooth execution of the workshop through coordination, technical support, and participant engagement on July 2025.
NITS Workshop and labs :- Mechanical Engineering Workshop, Structural Engineering Lab, Fluid Mechanics Lab, Computer Aided Design & Lab.

Leadership Activities

Social

- Head, **Razzmatazz** – **Incandescence**, NIT Silchar, 2023
- Head, **School Genius** – **Tecnoesis**, NIT Silchar, 2022

[1] CDEEP IIT Bombay [2] Master's course NIT Silchar [3] IISc Bengaluru